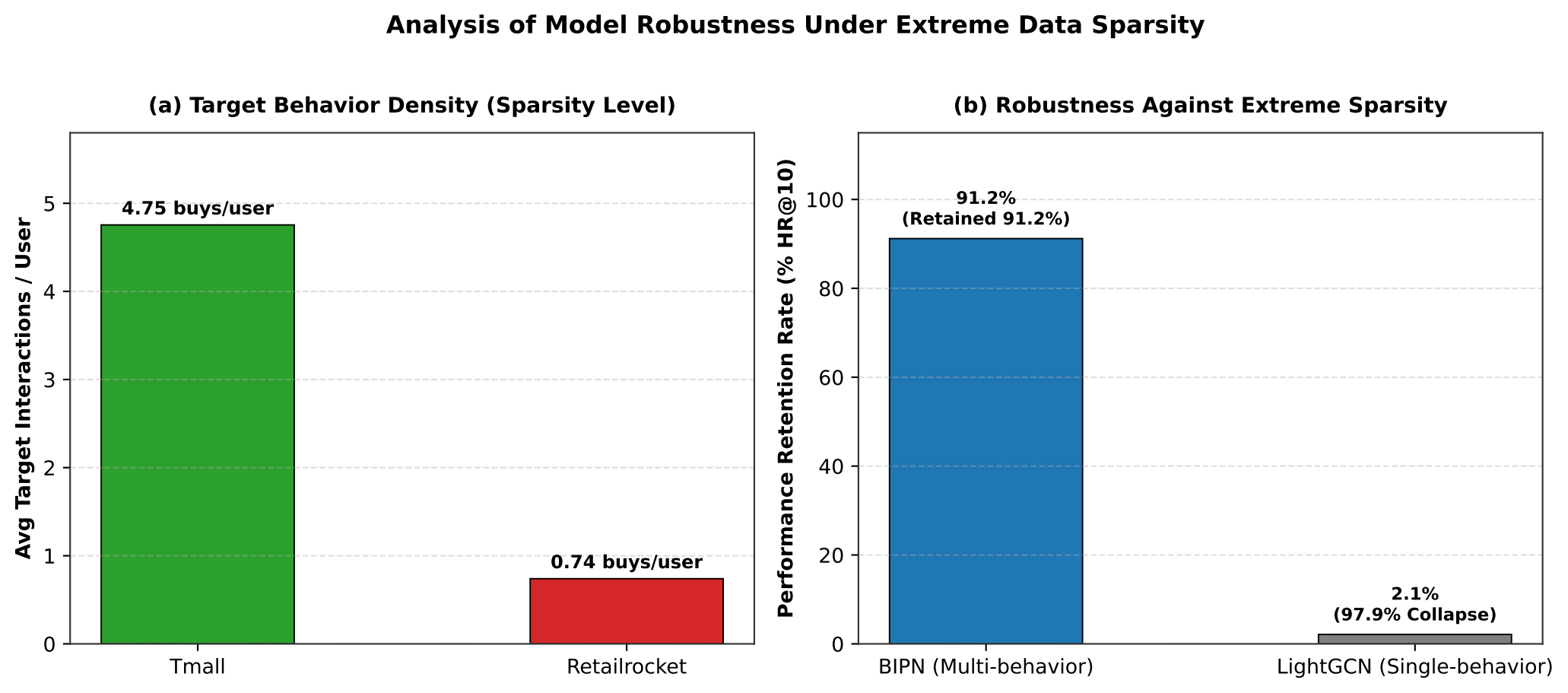
Analysis of Model Robustness Under Extreme Data Sparsity
(a) Target Behavior Density (Sparsity Level)
Avg Target Interactions / User
4.75 buys/user
0.74 buys/user
Tmall
Retailrocket
(b) Robustness Against Extreme Sparsity
Performance Retention Rate (% HR@10)
91.2%
(Retained 91.2%)
2.1%
(97.9% Collapse)
BIPN (Multi-behavior)
LightGCN (Single-behavior)